Do Now · retrieval

Lesson 102 · Choosing RISC, CISC, GPU and Multicore Systems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does RISC primarily describe?

Q2

What does a GPU excel at?

Q3

Can a RISC CPU be multicore?

Q4

Graph edges A–B, A–C, B–D: BFS from A with B before C?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — RISC/CISC and CPU/GPU are not a single four-way classification: a processor's ISA, core organisation and accelerators are different design properties.

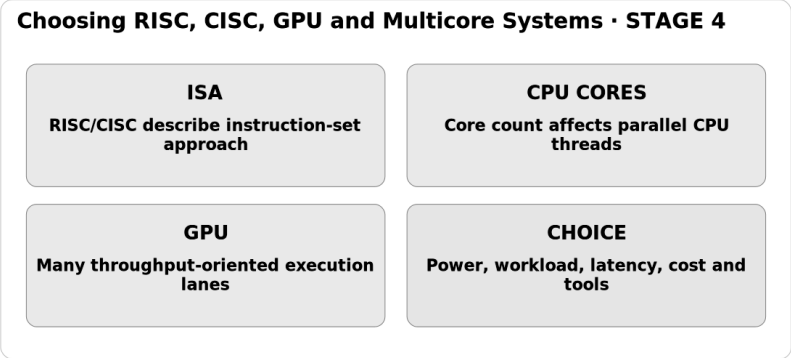
Core idea 2

Apply — A graphics or machine-learning workload with abundant parallel operations may benefit from a GPU; sequential/control-intensive jobs suit general CPU cores.

Core idea 3

Evaluate — Choose based on power, energy budget, latency, software compatibility, transfer overhead and ability to divide the workload; support claims with mechanism.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ISA — Instruction Set Architecture.

accelerator — Hardware specialised for particular workloads.

latency — Delay of a task.

throughput — Amount of work per unit time.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

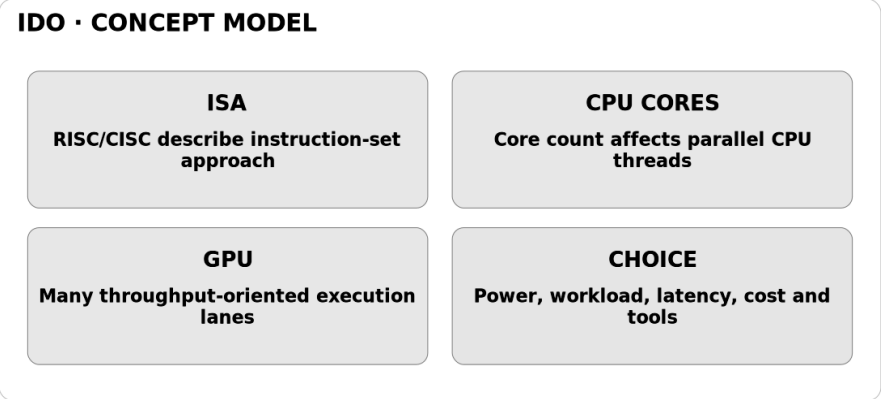
Problem

Recommend hardware for battery-powered sensor making a small control decision each second.

Follow the worked method

1. Explain the choice — A low-power embedded CPU (often using a RISC architecture) is reasonable; no large GPU is needed for tiny sequential work.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the choice
A GPU can accelerate similar independent pixel/shading operations, paired with a CPU for control.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain why RISC/CISC and multicore are not competing mutually exclusive categories. [6]

Question 2

Select suitable processor features for (a) small battery sensor (b) large parallel graphics processing. Justify. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate CPU vs GPU investment for a school's interactive office tasks and high-volume video editing. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every multicore processor must use a CISC instruction set.

Use key vocabulary from memory

1. Explain ISA in your own words.

2. Explain accelerator in your own words.

3. Explain latency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.